Internal Evaluation Metrics

Within cluster sum of squares (WCSS) (Inertia)

Given a dataset with k clusters, WCSS is defined as the sum of squared distances between each point and the centroid of the cluster to which belongs.

$$WCSS = \sum_{j=1}^{K} \sum_{x_i \in C_j} \| x_i - \mu_j \|^2$$

K : number of clusters

C_j : points in j^{th} cluster

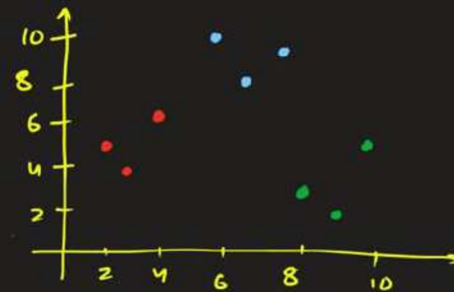
x_i : i^{th} point in the dataset

μ_j : centroid of the j^{th} cluster

$\| \cdot \|$: Euclidean distance

eg:

Cluster A	Cluster B	Cluster C
(2,5)	(8,3)	(6,10)
(3,4)	(9,2)	(7,8)
(4,6)	(10,5)	(8,9)



Calculate the within cluster square sum for the clustering given above.

Sol. step 1:

$$\mu_A = \left(\frac{2+3+4}{3}, \frac{5+4+6}{3}\right) = (3,5)$$

$$\mu_B = \left(\frac{8+9+10}{3}, \frac{3+2+5}{3}\right) = (9, 3.33)$$

$$\mu_C = \left(\frac{6+7+8}{3}, \frac{10+8+9}{3}\right) = (7,9)$$

Step 2:

$d^2(A1, \mu_A) = 1$, $d^2(A2, \mu_A) = 1$, $d^2(A3, \mu_A) = 2$

$d^2(B1, \mu_B) = 1.11$, $d^2(B2, \mu_B) = 1.77$, $d^2(B3, \mu_B) = 3.79$

$d^2(C1, \mu_C) = 2$, $d^2(C2, \mu_C) = 1$, $d^2(C3, \mu_C) = 1$

Step 3:

$$WCSS = 1+1+2+1.11+1.77+3.79+2+1+1 \simeq \boxed{14.67}$$

Dunn's Index

Aims to identify well-separated and compact clusters by comparing the inter-cluster distances to the intra-clusters distances. A higher dunn's index indicates better clustering.

$$D = \frac{\min_{i \neq j} d(C_i, C_j)}{\max_k d(C_k)}$$

$\leftarrow$ inter cluster distance

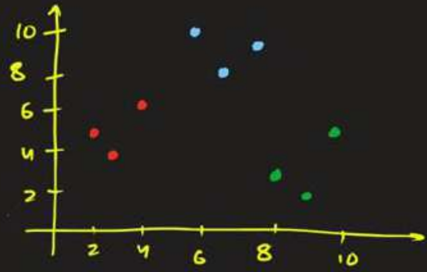
$\leftarrow$ intra cluster distance

$d(C_i, C_j)$ is the distance between clusters C_i and C_j

$d(C_k)$ is the diameter of cluster C_k

eg:

Cluster A	Cluster B	Cluster C
(2,5)	(8,3)	(6,10)
(3,4)	(9,2)	(7,8)
(4,6)	(10,5)	(8,9)



Calculate the Dunn's Index for the clustering given above.

Sol. Step 1: $\mu_A = \left(\frac{2+3+4}{3}, \frac{5+4+6}{3}\right) = (3,5)$

$\mu_B = \left(\frac{8+9+10}{3}, \frac{3+2+5}{3}\right) = (9, 3.33)$

$\mu_C = \left(\frac{6+7+8}{3}, \frac{10+8+9}{3}\right) = (7,9)$

$d_{AB} \simeq 6.23, \quad d_{BC} \simeq 6.012, \quad d_{AC} = 5.66$

Step 2: $d(A) = \max(d(A1, A2), d(A1, A3), d(A2, A3))$

$= \max(\sqrt{2}, \sqrt{5}, \sqrt{5}) = \sqrt{5}$

$d(B) = \max(\sqrt{2}, \sqrt{8}, \sqrt{10}) = \sqrt{10}$

$d(C) = \max(\sqrt{5}, \sqrt{5}, \sqrt{2}) = \sqrt{5}$

Step 3: $D = \dfrac{\min(d_{AB}, d_{BC}, d_{CA})}{\max(d(A), d(B), d(C))} \simeq \dfrac{5.66}{\sqrt{10}} \simeq \boxed{1.79}$

External Evaluation Metrics

Compare the clustering results with ground truth labels, providing a more direct assessment of clustering performance.

Contingency Matrix

displays the frequency distribution of clusters and ground truth classes.

For a clustering task with K clusters (prediction) and C ground (actual) truth classes, the contingency matrix will will have K rows (for clusters / prediction) and C columns (for classes / actual ground truth). Each cell (i,j) represents the number of data points that are assigned to i^{th} cluster and j^{th} class.

eg.

Data Point	(predict) Cluster	(Actual) True Label
1	1	A
2	1	A
3	1	A
4	1	A
5	1	A
6	1	B
7	2	A
8	2	B

Data Point	Cluster	True Label
9	2	B
10	2	B
11	2	C
12	3	B
13	3	B
14	3	C
15	3	C
16	3	C

Create the contingency table for the dataset given.

Sol. Cluster 1: 5 points from class A, 1 from class B

Cluster 2: 1 point from class A, 3 from class B, 1 from class C

Cluster 3: 2 points from class B, 3 from class C

	class A	class B	class C
Cluster 1	5	1	0
cluster 2	1	3	1
Cluster 3	0	2	3

Purity

measures the extent to which the clusters contain a single class.

It is the ratio of correctly assigned instances to the total number of instances.

$$\text{Purity} = \frac{1}{N} \sum_{j=1}^{C} \max_{i} |C_i \cap T_j|$$

N: Total number of data points

$|C_i|$: # instances in i^{th} cluster

$|T_j|$: # instances in j^{th} true class.

$$\text{Purity}_i = \frac{\max_{j} |C_i \cap T_j|}{|C_i|} \quad \text{(Purity of } i^{th} \text{ cluster)}$$

eg.

	class A	class B	class C	row Sum
cluster 1	5	1	0	6
cluster 2	1	3	1	5
cluster 3	0	2	3	5
Column sum	6	6	4	16

Calculate the purity of all the clusters and the overall purity of clustering algorithm.

Sol.

purity of cluster 1 $= \frac{5}{6} \simeq 0.83$

Purity of cluster 2 $= \frac{3}{5} = 0.6$

Purity of cluster 3 $= \frac{3}{5} = 0.6$

Overall purity $= \frac{1}{16}(5+3+3) = \frac{11}{16} = \boxed{0.6875}$

Rand Index (RI)

A measure of similarity between two data clusterings, calculated based on pairs of data points.

$$\text{Rand Index} = \frac{\text{Number of agreements}}{\text{Total number of pairs}}$$

Data Point	(predict) Cluster	(Actual) True Label
1	1	A
2	1	A
3	1	A
4	1	A
5	1	A
6	1	B
7	2	A
8	2	B

Data Point	Cluster	True Label
9	2	B
10	2	B
11	2	C
12	3	B
13	3	B
14	3	C
15	3	C
16	3	C

16 data points

$^{16}C_2 = \frac{16 \times 15}{2} = 120$

1,2 , 1, 14, 14, 16 ...

Agree
$\underline{\underline{1, 2}} \longrightarrow$ if they are in the same cluster, they should be in the same true class (TP)

or if they are in the different cluster, they should be

$\underline{\underline{3, 12}} \rightarrow$ in different true class (TN)

Disagreement
$\underline{\underline{5, 6}} \Rightarrow$ belong to same cluster but different true class (FP)

$\underline{\underline{8, 13}} \Rightarrow$ belong to different clusters but same true class (FN)

$$\text{Rand Index} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{TP + TN}{{}^{N}C_2}$$

2

eg

	class A	class B	class C	row sum
Cluster 1	5	1	0	6
cluster 2	1	3	1	5
cluster 3	0	2	3	5
Column sum	6	6	4	16

Calculate the rand index for the clustering dataset given.

Sol.

$$TP = \binom{5}{2} + \binom{3}{2} + \left[\binom{3}{2} + \binom{2}{2}\right] = 10 + 3 + 4 = 17$$

FP for cluster 1

Cluster 1
Cluster 1 } Classes are different

$$FP = (5\times1 + 5\times0 + 1\times0) + (1\times3 + 3\times1 + 1\times1) + (0\times2 + 2\times3 + 0\times3) = 5 + 7 + 6 = 18$$

FN for class A

Class A
Class A } clusters are different

$$FN = (5\times1 + 1\times0 + 5\times0) + (3\times1 + 3\times2 + 2\times1) + (0\times1 + 1\times3 + 3\times0)$$

$$= 5 + 11 + 3 = 19$$

$$TN = \binom{n}{2} - TP - FP - FN = \binom{16}{2} - 17 - 18 - 19 = 66$$

$$\text{Rand Index} = \frac{TP + TN}{{}^{n}C_2} = \frac{17+66}{120} = \frac{83}{120} \simeq \boxed{0.6917}$$

Adjusted Rand Index (ARI)

observed agreement

expected agreement

$$\mathrm{ARI} = \frac{\sum_{ij}\binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \cdot \sum_j \binom{b_j}{2}\right] / \binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2} \cdot \sum_j \binom{b_j}{2}\right] / \binom{n}{2}}$$

expected (TP+TN)

max agreement

expected agreement

*

(proof discussed later)

eg

	class A	class B	class C	row Sum
cluster 1	5	1	0	6 ✓
cluster 2	1	3	1	5 ✓
cluster 3	0	2	3	5 ✓
Column sum	6	6	4	16

Calculate the adjusted rand index for the clustering dataset given.

Sol.

$$\text{Expected agreement} = \left[\binom{6}{2}+\binom{5}{2}+\binom{5}{2}\right]\left[\binom{6}{2}+\binom{6}{2}+\binom{4}{2}\right] \Big/ \binom{16}{2}$$

$$= \frac{35 \times 36}{120} = 10.5$$

$$\text{ARI} = \frac{\binom{5}{2}+\binom{3}{2}+\binom{2}{2}+\binom{3}{2} - 10.5}{\left(\frac{35+36}{2}\right) - 10.5} = \frac{17-10.5}{35.5-10.5} = \frac{6.5}{25} = \boxed{0.26}$$

=x=